B.Sc. Engg. CSE 6th Semester | 29 June 2026 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION | **SUMMER SEMESTER, 2024-2025**

DURATION: 1 HOUR 30 MINUTES | **FULL MARKS: 75**

CSE 4619: Peripherals and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

---

1. a) Explain the memory-mapped I/O concept by considering a practical scenario where a Keyboard is assigned to address `0080H` for input data acquisition and an LCD display is assigned to address `0082H` for output operation. Describe how a microprocessor accesses these devices and analyze the advantages and limitations of this approach. 10 (CO1) (PO1)

   b) Describe the major stages involved in the A/D conversion process and discuss the importance of sampling rate, quantization level, resolution, and reference voltage in determining the accuracy and performance of an ADC system. 8 (CO1) (PO1)

   c) Briefly differentiate between the concepts of Computer Systems, Embedded Systems, and Cyber Physical Systems. 7 (CO1) (PO1)

2. a) Discuss how the architecture, conversion technique, sampling rate, resolution, noise performance, and hardware complexity of Flash and Delta-Sigma ADC influence their suitability. 10 (CO3) (PO1)

   b) A temperature monitoring system uses an ADC with an input range of $V_{min} = 0$ Volt and $V_{max} = 20$ Volts. If the system requires a voltage resolution of 25 mV, determine the minimum number of bits required for the ADC. 8 (CO4) (PO2)

   c) What are the limitations of a binary weighted resistor DAC? A 6-bit D/A converter produces $V_{out} = 3$ for the digital input `111000`. Find the output voltage for the input `010100`. 7 (CO2) (PO2)

3. a) With appropriate examples differentiate between the concepts of PPI and PIC with respect to CPU operation handling. 9 (CO4) (PO2)

   b) If the system continuously reads switch states from Port A and displays the inverted pattern on the LEDs through Port B, where, Port C is used for handshaking signals; then, using 8155's control register contents explain how the ports of the 8155 should be configured. 8 (CO2) (PO2)

   c) What will be the Port C's pin functionalities when an external device connected to Port A expects bidirectional communication with handshaking? If Port B is also connected with an input device, then explain your reasoning with reference to the characteristics of Mode 0, Mode 1 and Mode 2 both for the Port A and Port B. 8 (CO1) (PO1)